

AWS Solutions Architect - Interview Preparation Notes

Cloud Computing

Cloud computing is the delivery of computing resources over the internet. Service Models: IaaS, PaaS, SaaS. Deployment Models: Public, Private, Hybrid, Multi-cloud. Benefits: Scalability, elasticity, pay-as-you-go pricing, global reach, high availability. Interview Questions: Difference between scalability and elasticity, CAPEX vs OPEX.

AWS Global Infrastructure

Regions, Availability Zones, Edge Locations, Local Zones, Wavelength Zones. Design applications across multiple AZs for high availability.

EC2

Amazon EC2 provides virtual servers. Topics: AMI, instance types, security groups, key pairs, Elastic IPs, placement groups, instance lifecycle, Auto Scaling. Compare On-Demand, Reserved, Spot, and Dedicated Hosts.

Storage Services

EBS: Block storage for EC2. EFS: Shared file storage. S3: Object storage with storage classes, versioning, lifecycle policies, replication, encryption, static website hosting. FSx: Managed file systems.

Networking

VPC, CIDR, Subnets, Route Tables, Internet Gateway, NAT Gateway, Security Groups, NACLs, VPC Peering, VPN, Direct Connect.

Databases

RDS, DynamoDB, ElastiCache, Redshift. Multi-AZ vs Read Replica. SQL vs NoSQL.

IAM & Security

IAM Users, Groups, Roles, Policies, MFA. Principle of Least Privilege. KMS, CloudTrail, GuardDuty, AWS Config.

Route 53

DNS service. Hosted Zones, Record Types, Routing Policies: Simple, Weighted, Latency, Failover, Geolocation.

Monitoring

CloudWatch Metrics, Logs, Alarms, Dashboards. CloudTrail for auditing. Trusted Advisor recommendations.

Application Integration

SNS, SQS, EventBridge. Difference between SNS and SQS is a common interview topic.

Load Balancing & Auto Scaling

ALB, NLB, GWLB. Dynamic scaling, scheduled scaling, target tracking.

Serverless

AWS Lambda basics, event-driven architecture, integrations with S3, API Gateway, SNS.

Interview Tips

Explain architecture decisions, security best practices, high availability design, disaster recovery strategies, cost optimization, and Well-Architected Framework pillars.

Important AWS Interview Questions

1. Difference between Security Group and NACL.
2. Difference between EBS, EFS, and S3.
3. What is an AMI?
4. Explain Multi-AZ vs Read Replica.
5. What is Auto Scaling?
6. Explain Route 53 routing policies.
7. What is IAM Role?
8. Explain VPC components.
9. What is CloudFront?
10. Explain AWS Well-Architected Framework.